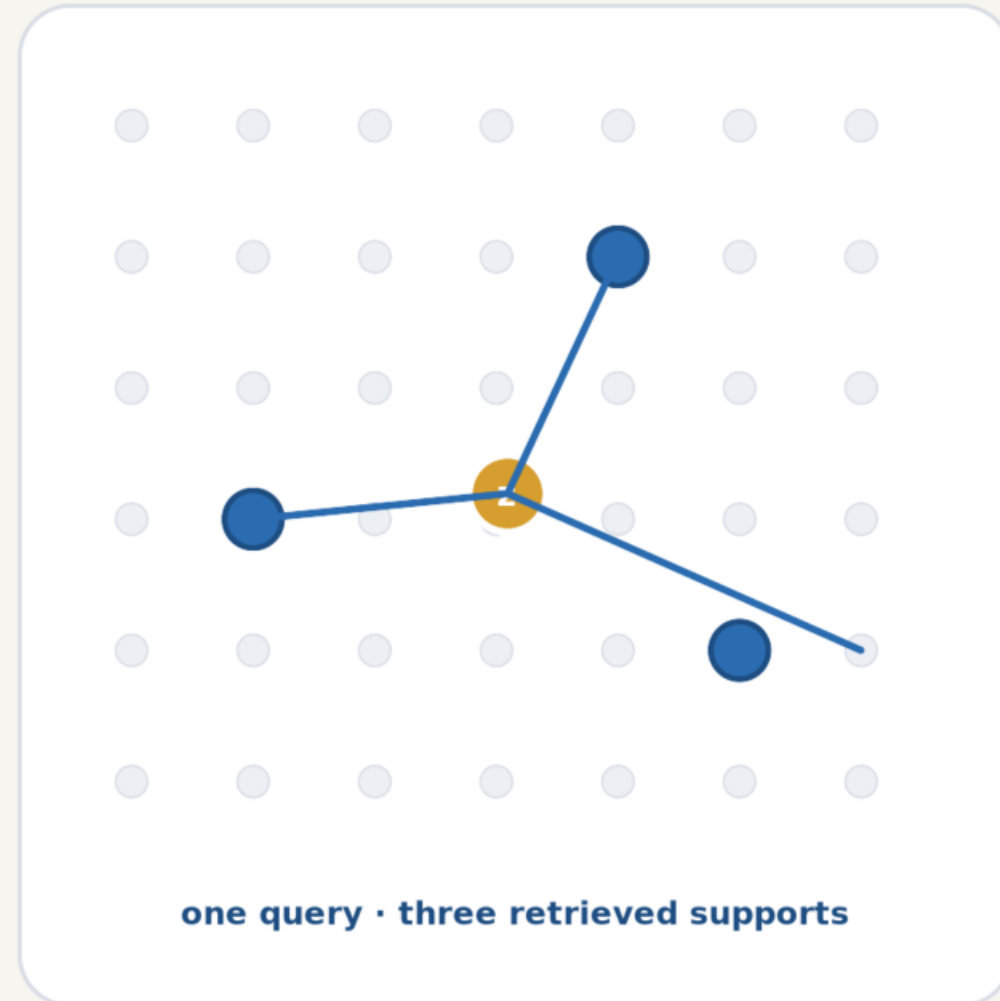


Sparse Separable Memory for Sequence Models

Visual intuition, anchored to the operation each equation performs.

capacity $\propto \frac{M}{K}$ active experts $\propto K$



The technical thesis: capacity grows with M, active experts with

MEMORY BANK · M SUPPORTS



Only $K = 3$ experts are active for this query

stored expert parameters $\propto M$
active expert FLOPs $\propto K$
 $K \quad M$

exact routing: $O(M d_z)$
product / ANN: sublinear

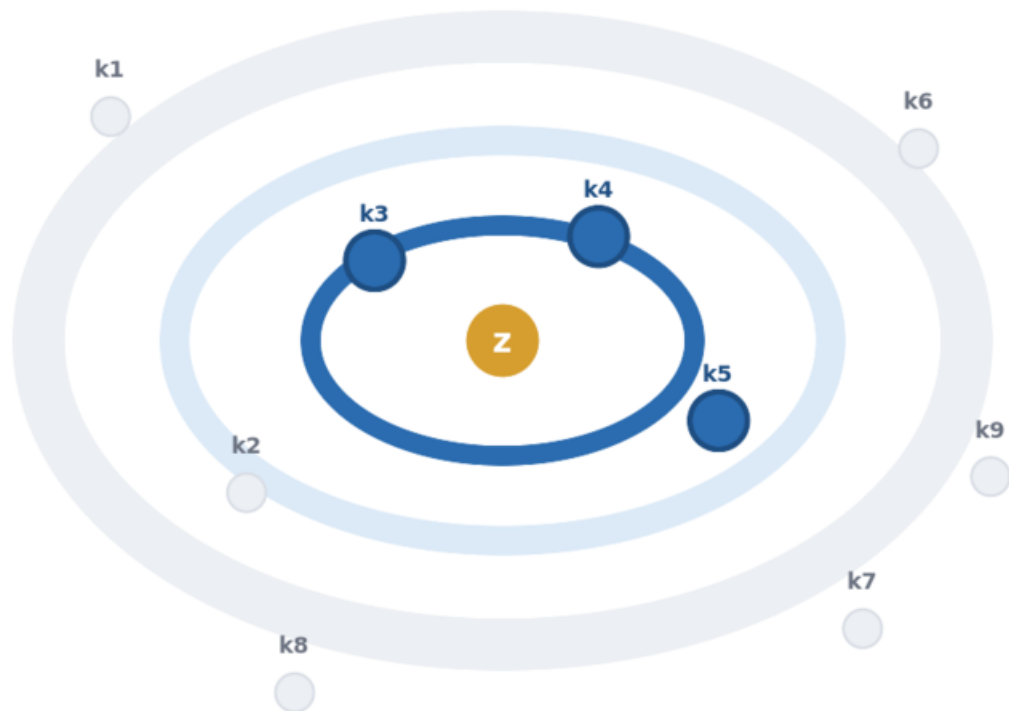
Sparse experts do not by themselves
solve routing cost.

Technical claim

Capacity and active expert work can decouple
if the router also scales.

Routing is kernel localization

KEY SPACE



D shapes the contours · τ controls sharpness · Top-K selects support

$$s_i(z) = -\frac{1}{2\tau} \frac{(z - k_i)^T D (z - k_i)}{(z - k_i)^T D (z - k_i)}$$

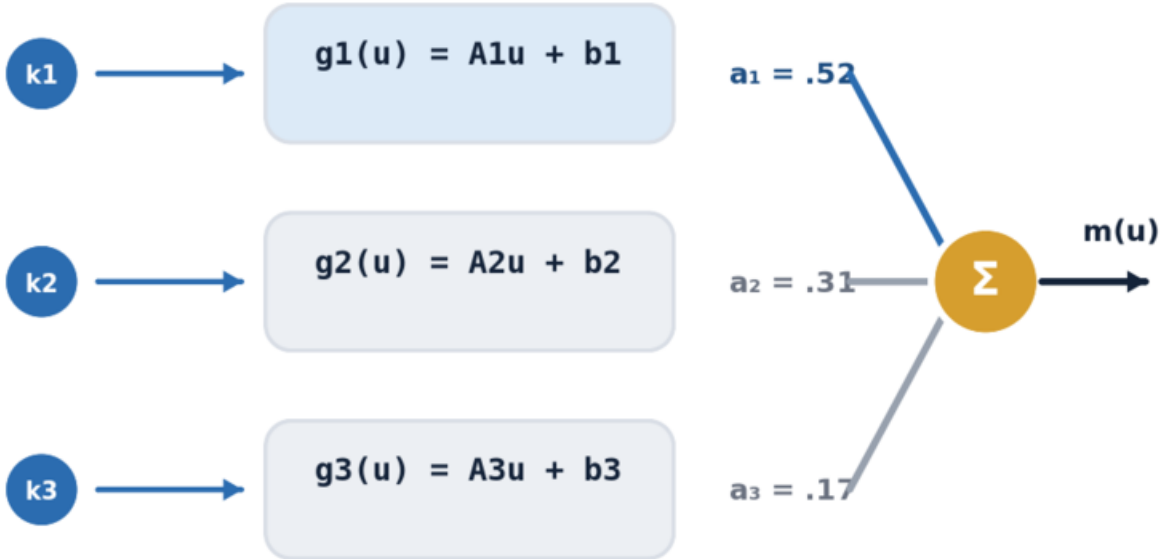
$$I(z) = \text{TopK}_i \ s_i(z)$$

$$a_i(z) = \frac{\exp(s_i)}{\sum_{j \in I(z)} \exp(s_j)}$$

The visual neighborhood is the metric defined by D —not a metaphorical cluster

Retrieval is a local model, then a gated residual

SELECTED LOCAL EXPERTS



$$g_i(u) = A_i u + b_i$$

$$m(u) = \sum_{i \in I(z)} a_i(z) g_i(u)$$

$$h^+ = h_{\text{base}}^+ + \gamma(u) W_o m(u)$$

What each term means

$I(z)$: selected neighborhood

A_i, b_i : local algebra

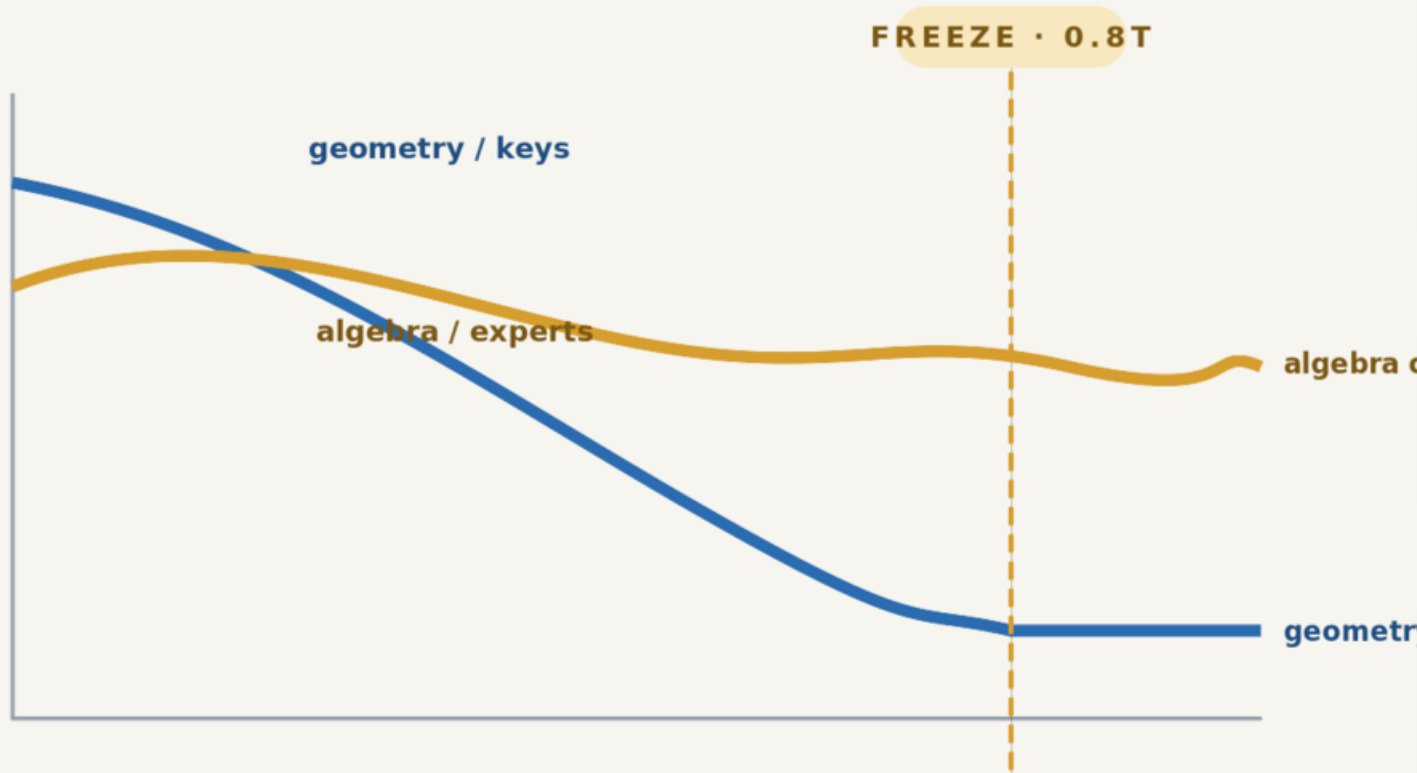
$\gamma(u)$: learned permission to use memory

The gate begins near zero, so the architecture begins close to its baseline.

Sparse Separable Memory

“Separable” means two parameter classes and two timescales

UPDATE MAGNITUDE



$$\theta_A \leftarrow \theta_A - \eta_A \nabla_{\{\theta_A\}} L$$

$$\theta_G \leftarrow \theta_G - \eta_G \nabla_{\{\theta_G\}} L$$

$$\eta_G \quad \eta_A$$

$$\eta_G(t) = 0, \quad t \geq 0.8T$$

$$\|K_T - K_{\{0.8T\}}\|_2 \leq 10^{-10}$$

Learn the map → stabilize it → freeze it → tune against stable routing

Each research branch is a different mathematical intervention

T-KAM-F · FIXED GEOMETRY

$$k_i \leftarrow \text{data center}_i$$

Is useful geometry available without learning keys?

T-KAM-L · LEARNED GEOMETRY

$$k_i = k_i^0 + \Delta k_i$$

Does support placement improve without instability?

T-KAM-ALT · ALTERNATING

$$\theta_A \text{ step} \leftrightarrow \theta_G \text{ step}$$

Does timescale separation improve optimization?

ONLINE ALGEBRA

$$\theta_{\{A, t+1\}} = \theta_{\{A, t\}} - \eta_{\text{on}} \nabla_{\theta} \mathcal{L}$$

Can local experts recover quickly after a shift?

DUAL MEMORY

$$m_t = g_t m_t^G + (1 - g_t) m_t^E$$

Do persistent and recent memory complement each other?

PRODUCT-KEY ROUTING

$$k_{\{pq\}} = k_p^1 \oplus k_q^2$$

Can retrieval scale without exact search?

Evidence boundary

A result for one intervention cannot validate another intervention.

Confirmation is a paired test of a prespecified estimand

156

fixed rows

30

primary pairs

24

replication pairs

50M

tokens / row

PRIMARY + REPLICATION

both must pass

$$\delta_s = \log(L_{\text{KAM-F},s} / L_{\text{T-WIDE},s})$$

Pairing removes seed-level nuisance variation:
each architecture sees the same training and data-order seed.

The interval must clear a 2% improvement—
not merely exclude zero.

$$p_{\text{sign-flip}} \leq 0.05$$

AND

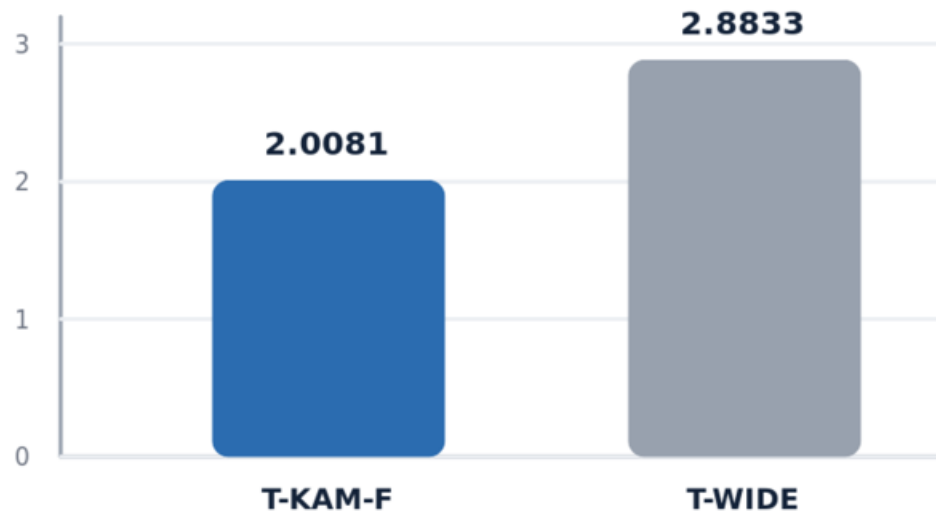
$$CI_{95\%}, \text{upper}(\text{mean } \delta) \leq \log(0.98)$$

Promotion rule**PrimaryPass $\wedge$ ReplicationPass**

Intermediate evidence: large point estimate, insufficient information

Pilot validation loss at registered checkpoint

Lower is better · $n = 3$ paired seeds



$$2.0081 / 2.8833 - 1 = -0.304 \rightarrow 30.4\% \text{ lower}$$

What the pilot can—and cannot—say

DIRECTIONAL ONLY

exact paired permutation $p = 0.25$

Holm-adjusted $p = 0.75$

The magnitude motivates confirmation;
the information is insufficient for a claim.

CONFIRMATION STATUS · JUL 28, 6:40 PM EDT

43 / 156 complete

4 running · 109 pending · 0 failures

Partial scientific effects remain uninspected.

Promising enough to confirm; not strong enough to conclude.

The decision decomposes into performance and mechanism gate

$\text{AccelerateFixed} = \text{PrimaryPass} \wedge \text{ReplicationPass}$

$\text{ClaimLearned} = \text{AccelerateFixed} \wedge \text{LifecyclePass}$
 $\wedge \text{LearnedVsFixedPass}$



FIXED KAM PASSES

Accelerate fixed / data-centered geometry.
Scale banks and language settings.



FIXED PASSES; LEARNED FAILS

Retain stable geometry.
Reject learned-memory claims.



PRIMARY OR REPLICATION FAILS

Reject a general quality advantage.
Test only separately supported niches.

Questions for discussion

- Is 2% the right scientific margin?
- Should scale-up optimize quality, active compute, or adaptation speed?
- Which mechanism result would most change the direction?

**Narrow claim, explicit gate,
mechanism-specific evidence.**